VERSION 1

UNIVERSITY OF TORONTO

FACULTY OF APPLIED SCIENCE AND ENGINEERING

Multiple Choice Assessment
Due: October 29^{th} 2020 by 2359 Eastern Daylight Time (Toronto)
First Year
APS110 - INTRODUCTORY CHEMISTRY AND MATERIALS SCIENCE
Exam Type: X (Open Book; Internet, if not communicated/posted during test; no communication with others)
Examiners: M Kortschot, CQ Jia, SD Ramsay

*Note: as this is a final assessment, late submissions will not be accepted. This means that if you do not submit your responses successfully to Crowdmark by the deadline you will not be able to submit and you will need to submit an official petition. Therefore, it is highly advisable that you make plans to complete and **submit your final responses 2-3 hours early.***

Instructions:

- You must correctly indicate your test version code on your the Microsoft form. If you do not, or fill this in incorrectly, your final assessment will not be graded.
- Answer all questions.
- Fill in the answers on the Microsoft Form.

1. Which of the following are the correct units for strain?

   (a) Dimensionless
   (b) mm
   (c) m
   (d) inches

2. The Abraham Lincoln Bridge, spanning the Ohio River from Louisville, Kentucky to Jeffersonville, Indiana, is designed with expansion joints to provide for 1.04 m of linear thermal expansion along the length of the bridge. Consider if this bridge had been built without these expansion joints and 1.04 m of linear expansion occurred, but the bridge was rigidly fixed at both ends, preventing this expansion. If the elastic modulus of the concrete in the bridge is 40 GPa and the compressive fracture strength of the concrete is 28 MPa, what stress would you expect to be imposed on the bridge?

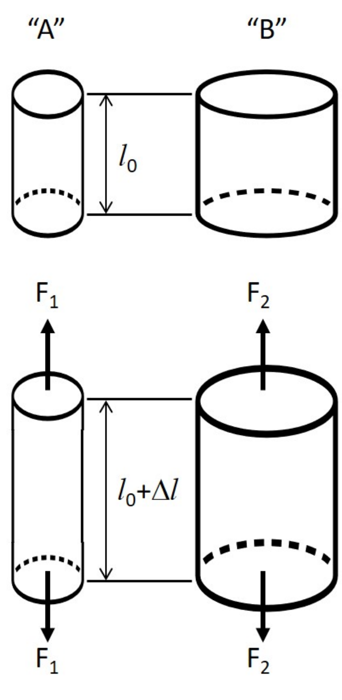
   (Expansion joint data from www.rjwatson.com/Projects/ohio-river-bridge-rjw-finger-joints, accessed October 25, 2020. Image from Censusdata at English Wikipedia, commons.wikimedia.org/wiki/File:2016WIKI_AbrahamLincolnBridgeLouKYJune12.jpg, CC BY-SA 4.0 creativecommons.org/licenses/by-sa/4.0, via Wikimedia Commons.))

   (a) 63 MPa
   (b) 21 MPa
   (c) 3 MPa
   (d) 6 GPa

3. Two ropes are cut, the first to 1 m in length and the second to 3 m in length. Both pieces of rope are then stretched by 2 cm. What will the ratio of the strain in the first to the strain in the second be?

   (a) 3
   (b) Equal because strain is sample size independent
   (c) 0.33
   (d) 2

4. Referring to the figure below, noting that both cylinders are made from the same material, which of the following statements are correct regarding cylinders "A" and "B"?



   (a) Both cylinders have the same stress and the same strain
   (b) Both cylinders have the same stress only
   (c) Both cylinders have the same strain only
   (d) Impossible to determine from information provided

5. A sample of a heavily cold worked 316L stainless steel is heated for several hours in a furnace at 1065 °C. The sample is then allowed to cool to room temperature. Relative to the value before the heat treatment, the Young's modulus after this heat treatment will be which of the following?

   (a) The same as
   (b) Greater than
   (c) Less than
   (d) Unable to determine with information provided

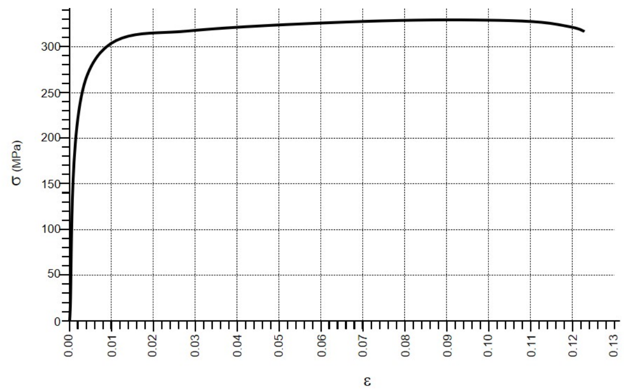
6. A sample of a heavily cold worked 316L stainless steel is heated for several hours in a furnace at 1065 °C. The sample is then allowed to cool to room temperature. Relative to the value before the heat treatment, the yield strength after this heat treatment will be which of the following?

   (a) Less than
   (b) Greater than
   (c) The same as
   (d) Unable to determine with information provided

7. A sample of a heavily cold worked 316L stainless steel is heated for several hours in a furnace at 1065 °C. The sample is then allowed to cool to room temperature. Relative to the value before the heat treatment, the plastic deformation at fracture after this heat treatment will be which of the following?

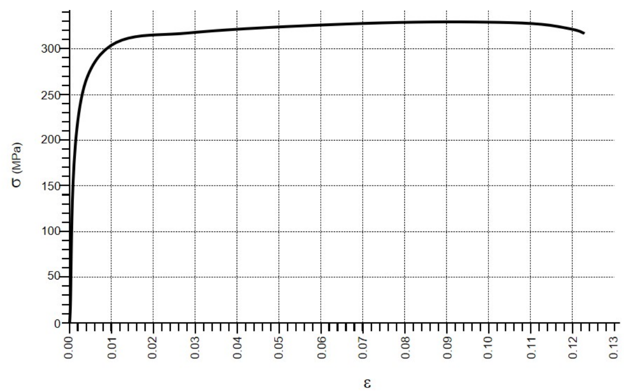
   (a) Greater than
   (b) Less than
   (c) The same as
   (d) Unable to determine with information provided

8. Referring to the stress-strain curve below, which of the following is closest to the value for the yield strength?



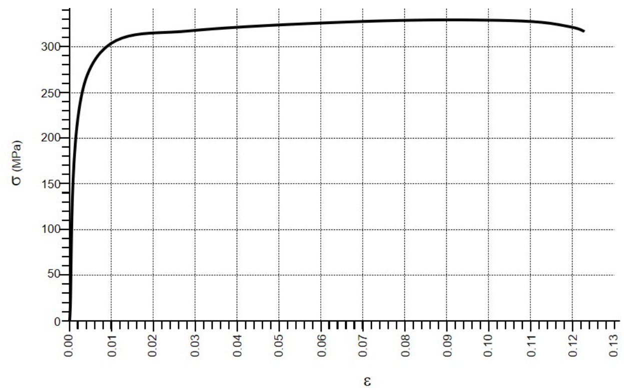
(a) 260 MPa
(b) 150 MPa
(c) 300 MPa
(d) None of these options.

9. Referring to the stress-strain curve below, which of the following is closest to the value for the ultimate tensile strength?



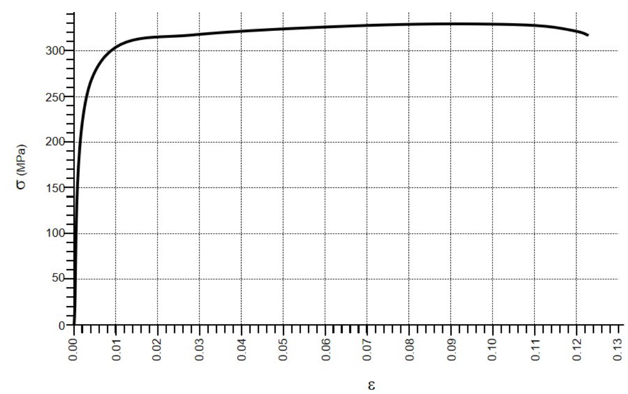
(a) 330 MPa
(b) 310 MPa
(c) 200 MPa
(d) 260 MPa

10. Referring to the stress-strain curve below, which of the following is closest to the value for the fracture strength?



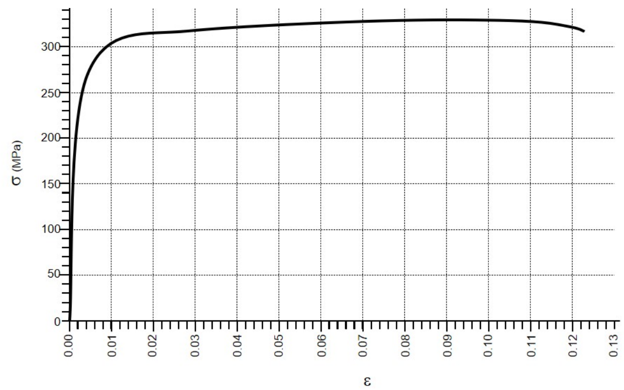
(a) 315 MPa
(b) 250 MPa
(c) 330 MPa
(d) 230 MPa

11. Referring to the stress-strain curve below, which of the following is closest to the value for the total strain at fracture?



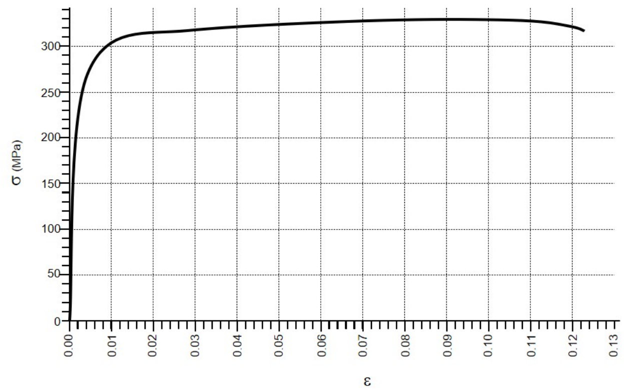
(a) 0.123
(b) 0.120
(c) 0.100
(d) 0.001

12. Referring to the stress-strain curve below, which of the following is closest to the value for the plastic strain at fracture?



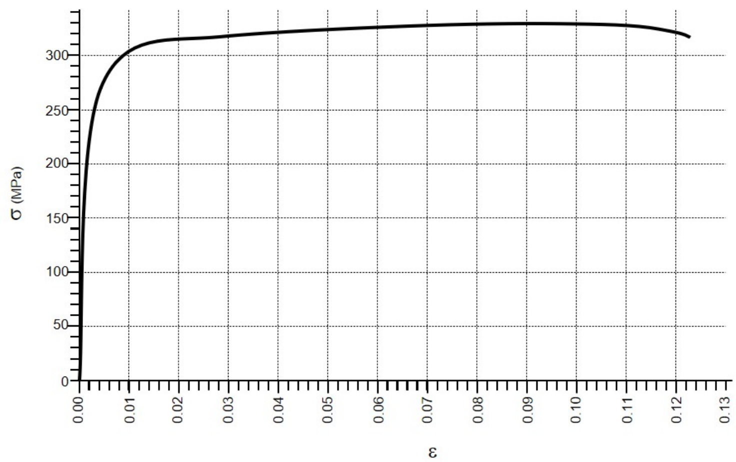
(a) 0.123
(b) 0.12
(c) 0.100
(d) 0.001

13. Referring to the stress-strain curve below, which of the following is closest to the value for the elastic strain recovered at fracture?



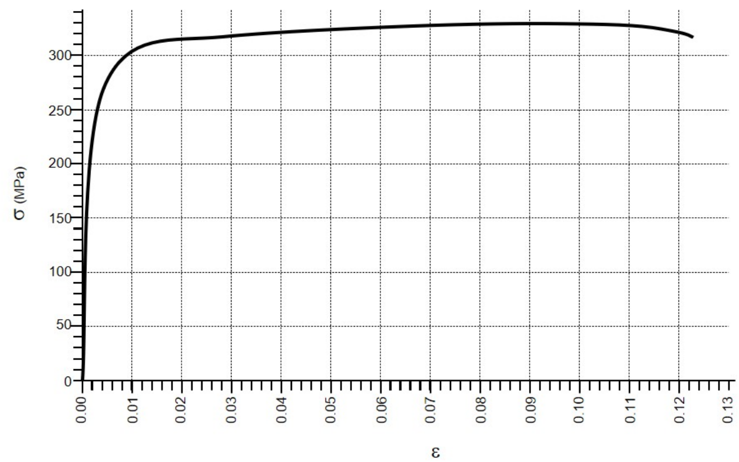
(a) 0.000
(b) 0.12
(c) 0.003
(d) 0.123

14. A sample of hypothetical metal having the stress-strain behaviour shown below is loaded to 280 MPa. If the sample has an initial length of 1 m, which of the following is closest to the length of this sample while this load is applied?



(a) 1.07 m
(b) 1 m
(c) 1.26 m
(d) 1.19 m

15. A sample of hypothetical metal having the stress-strain behaviour shown below is loaded to 280 MPa. If the sample has an initial length of 1 m, which of the following is closest to the length of this sample upon unloading?

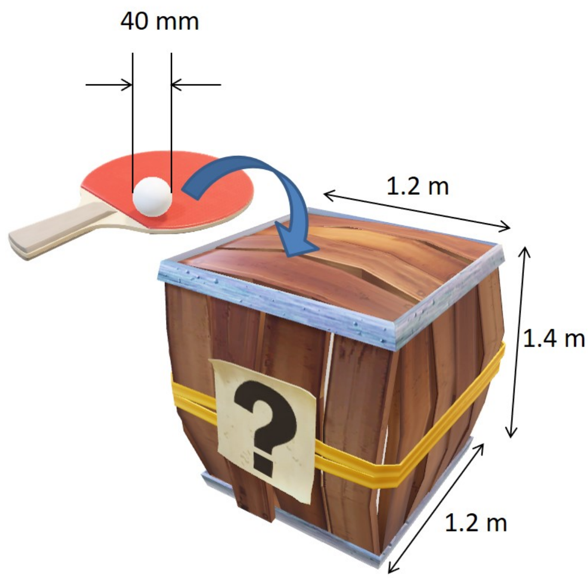


(a) 1.07 m
(b) 1 m
(c) 1.26 m
(d) 1.19 m

16. You have designed a lovely desk similar to the one shown in the figure below. The thickness is 82 mm, the width is 750 mm and the unsupported span is 1300 mm. You have made the desk from Poplar having a bending strength of 68 MPa. You are very protective of this desk, however, when you step away to eat a healthy meal, one of your family members stands in the middle of the unsupported span to reach up to hang a picture on the wall. If your family member weighs 63 kg will the desk break?

(a) No
(b) Yes
(c) Unable to determine from the information provided

17. A local restaurant is running a marketing promotion to increase business through their take-out business during the pandemic. They have filled a large box with table tennis balls and the challenge is to guess the number of balls in the box. To make this exciting, the restaurant has packed the table tennis balls as tightly as possible into the box. If the box has dimensions of $1.2\,\text{m} \times 1.4\,\text{m} \times 1.2\,\text{m}$, how many balls are in the box?



(a) 45, 000
(b) 890
(c) 41, 000
(d) 5, 600

18. Stress is described as being sample size independent for which of the following reasons?

(a) Because stress is force per unit area, so larger samples will also require larger force to achieve the same stress.
(b) Because stress is a dimensionless quantity
(c) Because stress is always the same, regardless of how many times you repeat a test
(d) None of these options.

19. Young's modulus is described as being structure independent for which of the following reasons?

    (a) Because Young's modulus relates to a specific bond type, which does not change with microstructural changes
    (b) Because Young's modulus is a dimensionless quantity
    (c) Because Young's modulus does not depend on the size of a sample being tested
    (d) None of these options.

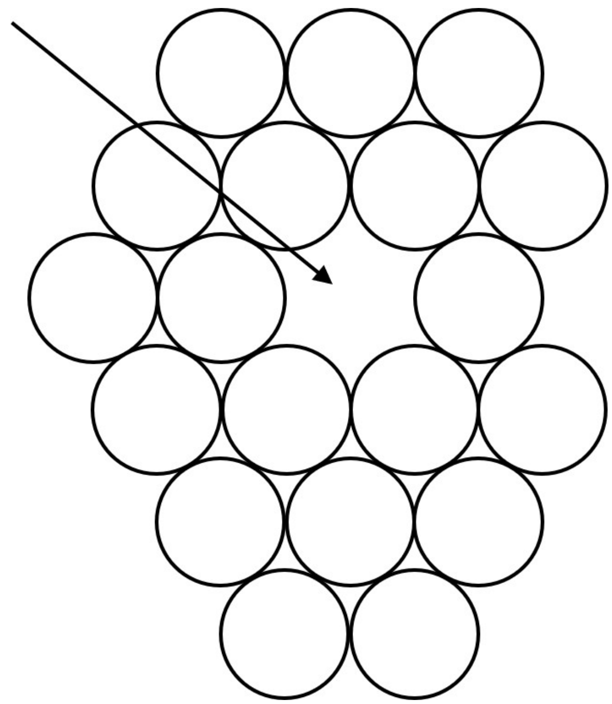
20. The stored elastic strain energy in tempered glass can be roughly related to which of the following?

    (a) The surface energy liberated during fracture
    (b) The strain energy surrounding dislocations in the glass
    (c) The thermal energy used to produce the glass
    (d) None of these options.

21. Which of the following is the best description of the difference between a vacancy and a pore?

    (a) A vacancy involves only a single missing atom, whereas a pore involves a massive number, say 10^9 of missing atoms
    (b) A vacancy occurs within a crystal, whereas a pore exists on the surface
    (c) A vacancy is a linear imperfection, whereas a pore is a three dimensional imperfection
    (d) None of these options.

22. The crystalline imperfection illustrated in the figure below is which of the following?

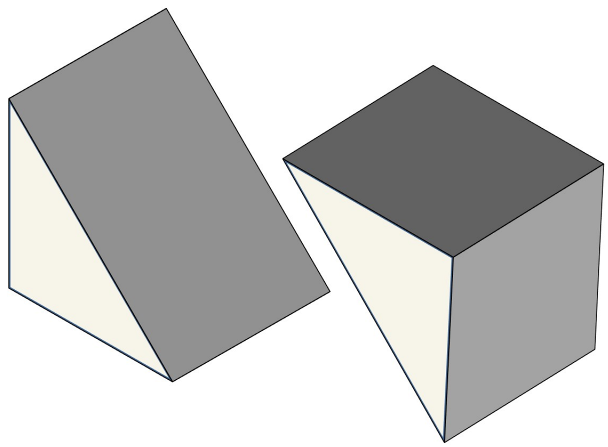


(a) Vacancy
(b) Grain boundary
(c) Interstitial atom
(d) Dislocation

23. Which of the following is the best explanation for the reason that a few atoms in a solid at room temperature may have enough kinetic energy to break free of their nearest neighbour atoms, while the majority are held in place by bonds?

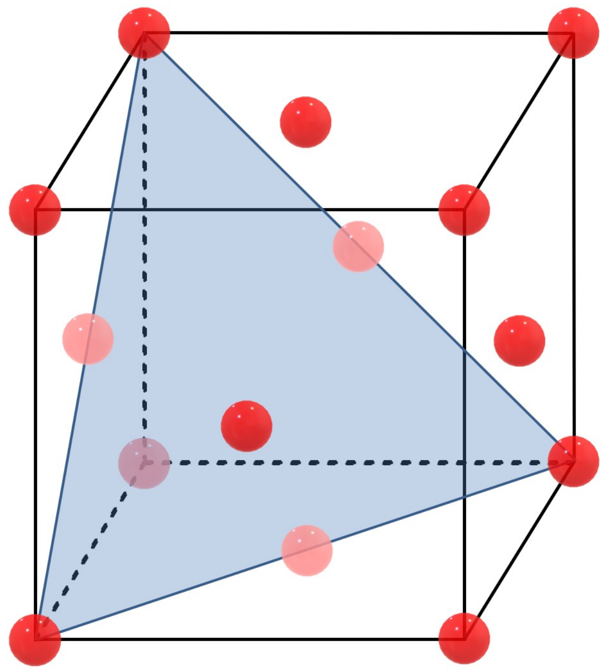
(a) There is a distribution of energies for a population of atoms
(b) Crystalline imperfections mean that there will randomly be some missing bonds
(c) This occurs only to impurity atoms
(d) These result from manufacturing defects

24. If a single unit of a BCC material was sliced as shown in the figure below, which of the following would correctly describe the fraction of the area of the sliced plane covered by atoms?



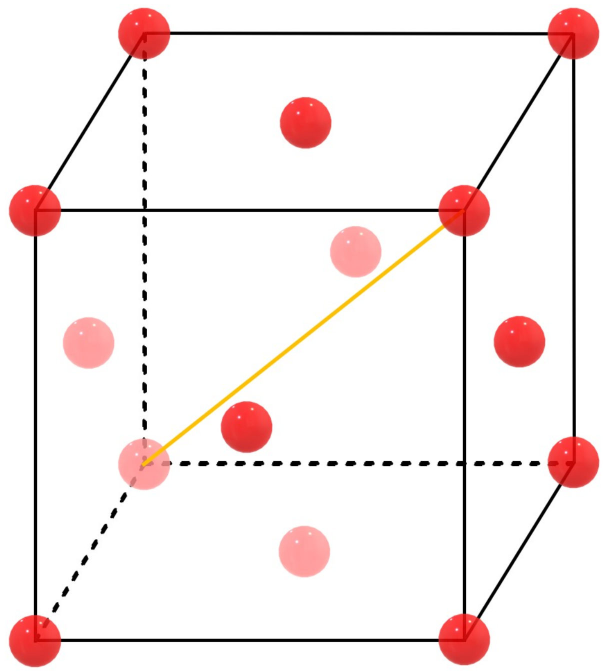
(a) $\frac{3\sqrt{2}\pi}{16}$
(b) $\frac{15\sqrt{2}\pi}{16}$
(c) $\frac{3\sqrt{2}\pi}{32}$
(d) None of these options.

25. Referring to the unit cell below, how many octahedral interstitial sites have centers residing on the blue plane?



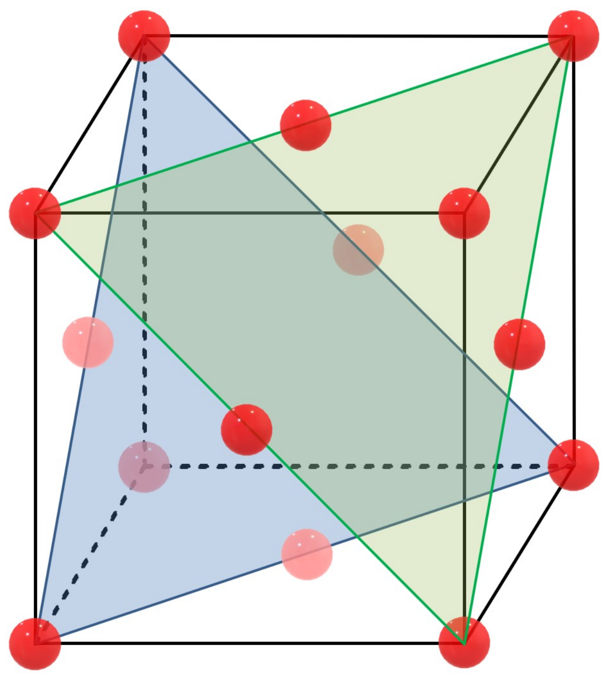
(a) 0
(b) 1
(c) 2
(d) 3

26. Referring to the unit cell below, how many octahedral interstitial sites have centers residing on the orange line?



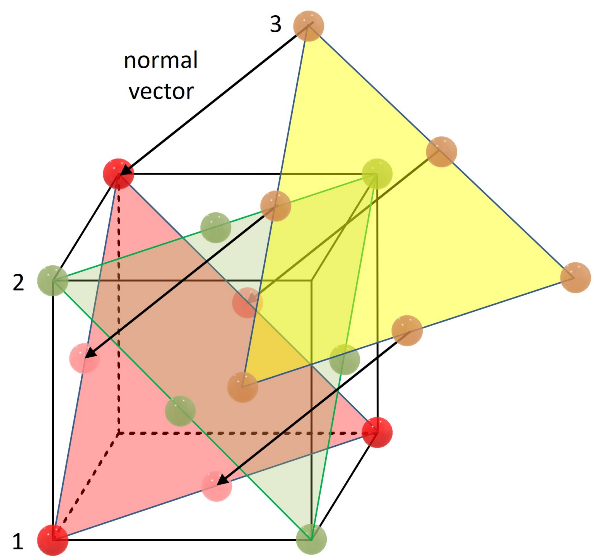
(a) 1
(b) 0
(c) 2
(d) 3

27. Referring to the unit cell below, how many octahedral interstitial sites have centers residing within the unit cell, exactly half way between the blue and green planes?



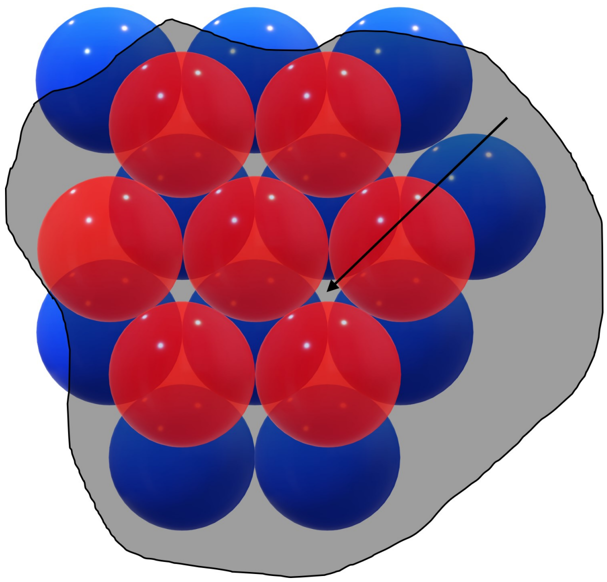
(a) 1
(b) 0
(c) 2
(d) 3

28. There's a lot going on in the figure below. Take a moment to work it out. The solid black directions are normal to the planes. Planes 1 and 2 have atoms residing in either cube corners or cube face-centers. Which crystal structure is illustrated here?



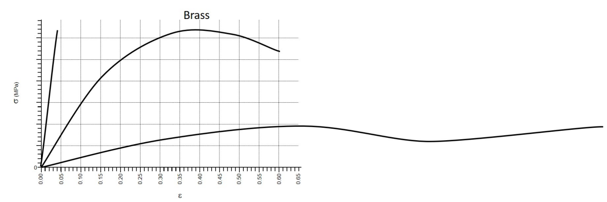
(a) HCP
(b) FCC
(c) BCC
(d) Simple Cubic

29. Two parallel planes of atoms are sketched below, viewed normal to the planes. The red plane is on top of the blue plane and a grey plane, with no atoms, is sketched and resides exactly half way between the red and blue planes. The arrow identifies a point on the grey plane. What is the coordination (with all atom colors) of the point indicated with the arrow?



(a) 6
(b) 4
(c) 3
(d) 12

30. Referring to the figure below, noting that one of the curves was from a brass sample tested in tension, which are the two materials that most likely produced the other two curves?



(a) Ceramic in compression, polymer
(b) Ceramic in tension, polymer
(c) Brittle polymer, ceramic in compression
(d) Brittle polymer, ductile polymer

Instructions:

- Congratulations, you've reached the end of the assessment!
- Remember to submit your answers to Microsoft Forms.